

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 2: Communication

Subtopic 2.1: Networks including the Internet

Concept 2.1.7: Cloud Computing (Public, Private)

Total Questions: 9

Mark Schemes begin on Page 5

Question 1 | Source: 9618_w25_qp_12 (Q8.a.i)

8 A restaurant accesses software and stores data using cloud computing.

(a) (i) Give **two** benefits of storing data using cloud computing.

- 1
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- 2
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[2]

Question 2 | Source: 9618_w25_qp_12 (Q8.a.ii)

8 A restaurant accesses software and stores data using cloud computing.

(ii) Give **two** drawbacks of using cloud computing.

- 1
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- 2
-

[2]

Question 3 | Source: 9618_s24_qp_13 (Q5.a.i)

5 A multimedia design company has an office with a LAN (local area network). The LAN can have up to 20 devices connected with cables and other devices connected using wireless access.

(a) The company has private cloud storage for its employees to store their work.

(i) Define the term **private cloud**.

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- [1]

Question 4 | Source: 9618_s24_qp_13 (Q5.a.ii)

5 A multimedia design company has an office with a LAN (local area network). The LAN can have up to 20 devices connected with cables and other devices connected using wireless access.

(a) The company has private cloud storage for its employees to store their work.

- (ii) Describe the benefits to the company of using private cloud storage instead of public cloud storage.

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..... [3]

Question 5 | Source: 9618_w23_qp_11 (Q2.c)

- 2 A school has a Local Area Network (LAN).

- (c) The students can save their school files on a public cloud.

Identify **two** drawbacks of the students storing their files on the public cloud.

1

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2

..... [2]

Question 6 | Source: 9618_w23_qp_13 (Q3.a.i)

- 3 A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

- (a) (i) Explain why the company uses a public cloud to store these courses.

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Question 7 | Source: 9618_w23_qp_13 (Q3.a.ii)

- 3 A company sells online Computer Science courses to students in different countries.

The courses are stored on a public cloud.

- (ii) Describe **two** disadvantages of storing data on a public cloud compared to storing data on a server in a Local Area Network (LAN).

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2

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[4]

Question 8 | Source: 9618_s21_qp_12 (Q5.c.i)

- 5 Seth uses a computer for work.

- (c) Seth accesses both software and data using cloud computing.

- (i) Give **two** benefits of storing data using cloud computing.

1

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2

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[2]

Question 9 | Source: 9618_s21_qp_12 (Q5.c.ii)

- 5 Seth uses a computer for work.

- (c) Seth accesses both software and data using cloud computing.

- (ii) Give **two** drawbacks of Seth using cloud computing.

1

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2

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[2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_12 (Q8.a.i)

8(a)(i)	1 mark per bullet point, max 2 marks • Cloud storage can be free of charge for a limited amount • There is no need for personal storage devices • The data can be accessed from any computer with an internet connection • Cloud data services usually include data backup and recovery services • It is easier to share data / collaborate with colleagues • Cloud storage capacity can be quickly and easily increased if needed	2
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Mark Scheme for Question 2 | Source: 9618_w25_ms_12 (Q8.a.ii)

8(a)(ii)	1 mark per bullet point, max 2 marks • The storage/cloud can only be accessed with an internet connection • There may be no backups // possibly no control over backups // reliance on third party for backups • Uploading / downloading data could take a long time • Potential issues with the cloud storage company impacting access e.g. downtime or technical problems • Potential compatibility issues if wanting to move the data • Limited storage availability without paying extra • ... meaning (possibly) more expensive in the long term • Possibly security issues	2
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Mark Scheme for Question 3 | Source: 9618_s24_ms_13 (Q5.a.i)

5(a)(i)	1 mark for: Dedicated/bespoke services/storage on a remote server only available to company	1
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Mark Scheme for Question 4 | Source: 9618_s24_ms_13 (Q5.a.ii)

5(a)(ii)	1 mark each to max 3 : e.g. • Not reliant on a third party • ... gives greater control over security/privacy • ... gives greater control over backup • Storage can be tailored/scalable to company requirements// an example e.g. the amount of storage accessible/ facilitating the sharing of files	3
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Mark Scheme for Question 5 | Source: 9618_w23_ms_11 (Q2.c)

2(c)	1 mark for each bullet point (max 2) - The students cannot access their files without a reliable internet connection - The amount of space for no payment may be limited so students will have to purchase more space if needed - The students do not have control over the backup (or security) of their work // the students are dependent on a third party for the (security and) backing up of their work	2
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Mark Scheme for Question 6 | Source: 9618_w23_ms_13 (Q3.a.i)

3(a)(i)	1 mark for each bullet point (max 2). - Courses must be available to anyone who wishes to follow them - Courses must be available on the internet - Company is willing to share infrastructure with other companies (public) ...which is more economic for the company	2
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Mark Scheme for Question 7 | Source: 9618_w23_ms_13 (Q3.a.ii)

3(a)(ii)	1 mark for each bullet point (max 2 for each disadvantage). - There could be a possible loss of control unlike the LAN ...because the data is stored on a remote infrastructure / someone else's infrastructure ...reliance on external agency to complete tasks, e.g. backups, security - Requires reliable internet connection ...to ensure access to the remote data, more likely with LAN - Increased recurring costs ...as cloud provider charges must be paid, costs for LAN once only.	4
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Mark Scheme for Question 8 | Source: 9618_s21_ms_12 (Q5.c.i)

5(c)(i)	1 mark per bullet point to max 2 - Cloud storage can be free (for small quantities) - No need for separate (high capacity) storage devices // saves storage on existing devices - Can access data from any computer with internet access - Most cloud data services will have in-built backup/disaster recovery - Security could be better - Can easily increase capacity - Data can be easily shared	2
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Mark Scheme for Question 9 | Source: 9618_s21_ms_12 (Q5.c.ii)

5(c)(ii)	1 mark per bullet point to max 2: • Can only access (the cloud) with internet access • Security may not be strong // no control over security • There may not be any backups // no control over backups • It can take a long time to upload/download the data • It can be more expensive in the long term • There could be a limit to the amount of storage unless paid for • There could be compatibility/access issues • There could be issues with the company offering cloud services	2
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